

Timothy James Wood

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First-class Biological Sciences graduate (University of Bristol) with two research projects in statistical population genetics under Prof. Mark Beaumont, now beginning an HDR UK-funded MSc in Bioinformatics at Bristol. Experienced in likelihood-based demographic inference (dadi, composite-likelihood model selection, Godambe uncertainty), genome-wide selection scans, and reproducible Python/R pipelines on HPC. Seeking a PhD in statistical genetics and machine learning, with a particular interest in the methodology of inference for genomic data.

EDUCATION

MSc Bioinformatics – University of Bristol

Sept 2026 – Sept 2027

- HDR UK MSc Scholarship (funded via Coeliac UK).
- Dissertation (supervisor: Prof. Mark Beaumont): simulation-based inference and population-genetic methods applied to the genetic architecture of coeliac disease.

BSc Biological Sciences, First Class Honours – University of Bristol

2023 – 2026

- Final-year research project 75% (1st); literature review 78% (1st).

RESEARCH EXPERIENCE

Research Assistant, Prof. Mark Beaumont's group – University of Bristol

Summer 2026

Composite-likelihood demographic inference for the Scottish wildcat and domestic cat, with Dr Dennis Prangle and Grace Yan (Schools of Biological Sciences and Mathematics)

- Built a folded joint site-frequency spectrum from 46 whole-genome cats (145k SNPs, 22.5 Mb callable sequence) and fitted three two-population demographic models in dadi, including an 11-parameter model with domestication and land-bridge epochs.
- Implemented model selection and uncertainty for composite likelihoods from first principles: composite-likelihood AIC and sandwich (Godambe) confidence intervals from 100 block-bootstrap replicates, with delta-method log-scale intervals and finite-difference stability checks.
- Diagnosed identifiability problems: showed that deep parameters lie on a single likelihood ridge (a “sloppy” model), that a nominal 12-parameter model has an effective count of ~293, and that a boundary optimum invalidates the best-scoring model, leading to a defensible primary result.
- Ran multi-round optimisation (200 restarts per model) as SLURM array jobs on Bristol's ACRC HPC; delivered a 30-page technical report and a documented GitHub repository (github.com/timj-wood/wildcat_models) intended as a baseline for later simulation-based inference work.

Final-year Research Project (BSc) – University of Bristol

Sept – Dec 2025

Genome-wide signals of selection and introgression in the European wildcat (supervisor: Prof. Mark Beaumont; data from Dr Jo Howard-McCombe, RZSS)

- Processed whole-genome SNP data for 108 wild, captive and domestic cats across 18 autosomes (VCFtools, PLINK; MAC, missingness, HWE and LD-thinning filters).
- Ran a PCA-based genome scan (pcadapt, $K = 5$) to identify 1,275 outlier intervals; annotated 11 candidate genes via g:Profiler, Ensembl and NCBI, with significant GO enrichment for GABA, taurine and calcium-ion transport.
- Developed paired population- and genotype-level PCA visualisations to separate introgression from demographic structure; critically evaluated drift vs. selection, linkage and threshold choices.

Literature Review (BSc) – Engineering predictability in living systems

Sept – Dec 2025

Supervisor: Prof. Thomas Gorochowski, Biocompute Lab, School of Biological Sciences

- Reviewed synthetic circuits, genome refactoring and recoding, AI-driven DBTL pipelines and biological foundation models (Evo, ESM-2, ProGen); argued that data-driven models capture evolutionary correlation rather than causal mechanism, and proposed context-aware data and adaptive feedback as remedies.

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, pandas, matplotlib), R (tidyverse, ggplot2), Bash/Unix, Git/GitHub, SLURM/HPC job arrays, LaTeX, advanced Excel.
Statistics	Maximum and composite likelihood, information criteria (AIC/CLAIIC), Godambe/sandwich estimators, block bootstrap, delta method, constrained optimisation (COBYLA/NLopt), PCA, multiple-testing correction, identifiability and sloppiness diagnostics.
Genomics	Site-frequency-spectrum and demographic inference (dadi), genome scans for selection (pcadapt), variant QC and filtering (VCFtools, PLINK), MSMC multihetsep format, functional annotation (g:Profiler, Ensembl, NCBI Gene), introgression and admixture analysis.

PROFESSIONAL EXPERIENCE

Audit Intern – PwC LLP, London

June – July 2025

- Supported the audit of WPP plc (FTSE 100) with a Senior Associate; trained in IFRS, audit methodology and proprietary data systems; produced review-ready analytical working papers from large financial datasets.

LEADERSHIP

President – University of Bristol Engineering Rugby Society

May 2025 – Apr 2026

- Ran club finances and Students' Union relations; secured £600 in sponsorship and £800 in event funding.

AWARDS

- HDR UK MSc Scholarship, 2026 (funded via Coeliac UK).
- First Class Honours, BSc Biological Sciences, University of Bristol, 2026.

REFEREES

Prof. Mark Beaumont – Professor of Statistics, School of Mathematics, University of Bristol. BSc, summer and MSc project supervisor.

Dr Dennis Prangle – Associate Professor in Statistics, School of Mathematics, University of Bristol. Summer project co-supervisor.

Prof. Thomas Gorochowski – Professor of Biological Engineering, School of Biological Sciences, University of Bristol. Literature review supervisor.

Contact details available on request.